Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If $(-4, 5)$ is one vertex and $7x - y + 8 = 0$ is one diagonal of a square, then the equation of second diagonal is

(A) $x + 3y = 21$

(B) $2x - 3y = 7$

(C) $x + 7y = 31$

(D) $2x + 3y = 21$

Q.2

A line through $\mathrm{A}(-5, -4)$ meets the line $\mathrm{x} + 3\mathrm{y} + 2 = 0, 2x + y + 4 = 0$ and $x - y - 5 = 0$ at B, C and D respectively. If $\left(\frac{15}{\mathrm{AB}}\right)^2 + \left(\frac{10}{\mathrm{AC}}\right)^2 = \left(\frac{6}{\mathrm{AD}}\right)^2$, then the equation of the line is

(A) $2x + 3y + 22 = 0$

(B) $5x - 4y + 7 = 0$

(C) $3x - 2y + 3 = 0$

(D) None of these

Q.3

The straight line $y = x - 2$ rotates about a point where it cuts the x-axis and becomes perpendicular to the straight line $ax + by + c = 0$. Then its equation is

(A) $ax + by + 2a = 0$

(B) $ax - by - 2a = 0$

(C) $bx + ay - 2b = 0$

(D) ay $-bx + 2b = 0$

Q.4

The distance of the line $2\mathrm{x} + \mathrm{y} = 3$ from the point $(-1, 3)$ in the direction whose slope is 1 is

(A) $\frac{2}{3}$

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(B) $\frac{\sqrt{2}}{3}$

(C) $\frac{2\sqrt{2}}{3}$

(D) $\frac{2\sqrt{5}}{3}$

Q.5

The equation of the straight line, the portion of which intercepted between the coordinate axes being divided by the point $(-5, 4)$ in the ratio $1:2$, is

(A) $8x + 5y = 60$

(B) $8x - 5y = 60$

(C) $-8x + 5y = 60$

(D) None of these

Q.6

A line which makes an acute angle θ with the positive direction of x -axis is drawn through the point $\mathrm{P}(3, 4)$ to meet the line $\mathrm{x} = 6$ at R and $\mathrm{y} = 8$ at S, then

(A) $\mathrm{PR} = 3\cos\theta$

(B) $\mathrm{PS} = -4\,\mathrm{cosec}\,\theta$

(C) $\mathrm{PR} - \mathrm{PS} = \frac{2(3\sin\theta + 4\cos\theta)}{\sin 2\theta}$

(D) $\frac{9}{(\mathrm{PR})^2} + \frac{16}{(\mathrm{PS})^2} = 1$

Q.7

Let (h, k) be a fixed point where $h > 0, k > 0$. A straight line passing through this point cuts the positive direction of the coordinate axes at the points P and Q. Then the minimum area of the $\Delta OPQ . O$ being the origin, is

(A) $4hk$ sq. units

(B) $2hk$ sq. units

(C) $3hk$ sq. units

(D) None of these

Q.8

The equation $\left(x^2 - a^2\right)^2 \left(x^2 - b^2\right)^2 + c^4 \left(y^2 - a^2\right)^2 = 0$
represents $(\mathrm{c} \neq 0)$

(A) 8 points

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(B) two circles

(C) 4 lines

(D) None of these

Q.9

The equation of the straight line which passes through the point $(-4, 3)$ such that the portion of the line between the axes is divided internally by the point in the ratio $5:3$ is

(A) $9x - 20y + 96 = 0$

(B) $9x + 20y = 24$

(C) $20x + 9y + 53 = 0$

(D) None of these

Q.10

D is a point on AC of the triangle with vertices $\mathrm{A}\ (2, 3), \mathrm{B}(1, -3), \mathrm{C}(-4, -7)$ and BD divides ABC into two triangles of equal area. The equation of the line drawn through B at right angles to BD is

(A) $y - 2x + 5 = 0$

(B) $2y - x + 5 = 0$

(C) $y + 2x - 5 = 0$

(D) $2y + x - 5 = 0$

Q.11

The length of the perpendicular from the origin to a line is 7 and line makes an angle of 150° with the positive direction of y -axis, then the equation of the line is

(A) $\sqrt{3}x + y = 7$

(B) $\sqrt{3}x - y = 14$

(C) $\sqrt{3}x + y + 14 = 0$

(D) $\sqrt{3}x + y - 14 = 0$

Q.12

A light ray emerging from the point source placed at $P(2, 3)$ is reflected at a point Q on the y -axis. It then passes through the point $R(5, 10)$. The coordinates of Q are

(A) $(0, 3)$

(B) $(0, 2)$

(C) $(0, 5)$

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(D) None of these

Q.13

A straight line L with negative slope passes through the point $(8, 2)$ and cuts the positive coordinate axes at points P and Q. as L varies the absolute minimum value of $OP + OQ$ is (O is origin)

(A) 28

(B) 15

(C) 18

(D) 10

Q.14

Consider the points $A(0, 1)$ and $B(2, 0)$, and P be point on the line $4x + 3y + 9 = 0$. The coordinates of P such that $|PA - PB|$ is maximum are

(A) $(-12/5, 17/5)$

(B) $(-84/5, 13/5)$

(C) $(-24/5, 17/5)$

(D) $(0, -3)$

Q.15

The point $A(2, 1)$ is translated parallel to the line $x - y = 3$ by a distance of 4 units. If the new position A' is in the third quadrant, then the coordinates of A' are

(A) $(2 + 2\sqrt{2}, 1 + 2\sqrt{2})$

(B) $(-2 + \sqrt{2}, -1 - 2\sqrt{2})$

(C) $(2 - 2\sqrt{2}, 1 - 2\sqrt{2})$

(D) None of these

Q.16

The equation of straight line passing through $(-a, 0)$ and making a triangle with the axes of area T is

(A) $2Tx + a^2y + 2aT = 0$

(B) $2Tx - a^2y + 2aT = 0$

(C) $2Tx - a^2y - 2aT = 0$

(D) None of these

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Q.17

What is the equation of the line which passes through $(4, -5)$ and is perpendicular to $3x + 4y + 5 = 0$?

(A) $4x - 3y - 31 = 0$

(B) $3x - 4y - 41 = 0$

(C) $4x + 3y - 1 = 0$

(D) $3x + 4y + 8 = 0$

Q.18

A plane parallel to y-axis passing through line of intersection of planes $x + y + z = 1$ and $2x + 3y - z - 4 = 0$ which of the point lie on the plane.

(A) $(3, 2, 1)$

(B) $(-3, 0, 1)$

(C) $(-3, 1, 1)$

(D) $(3, -1, 1)$

Q.19

Through the point $\mathrm{P}(\alpha, \beta)$, where $\alpha\beta > 0$, the straight line $\frac{x}{a} + \frac{y}{b} = 1$ is drawn so as to form with co-ordinate axes a triangle of area S. If $ab > 0$, the least value of S is

(A) $2\alpha\beta$

(B) $\alpha\beta$

(C) $3\alpha\beta$

(D) None of these

Q.20

An equation of a line through the point $(1, 2)$ whose distance from the point $(3, 1)$ has the greatest value is

(A) $y = 2x$

(B) $y = x + 1$

(C) $x + 2y = 5$

(D) $y = 3x - 1$

Answers & Solutions

Q.1 Answer:

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$x + 7y = 31$

Solution:

One vertex of square is $(-4, 5)$ and equation of one diagonal is $7x - y + 8 = 0$ Diagonal of a square are perpendicular and bisect each other Let the equation of the other diagonal be $y = mx + c$ where m is the slope of the line and c is the y -intercept. Since this line passes through $(-4, 5) \therefore \quad 5 = -4\,m + c \ldots$ (i)

Since this line is at right angle to the line $7x - y + 8 = 0$ or $y = 7x + 8$, having slope $= 7 \therefore$ $7 \times m = -1$ or $\quad m = \frac{-1}{7}$

Putting this value of m in equation (i) we get $c = 5 - \frac{4}{7} = \frac{31}{7}$

Hence equation of the other diagonal is $y = -\frac{1}{7}x + \frac{31}{7} \quad$ or $x + 7y = 31.$

Q.2 Answer:

$2x + 3y + 22 = 0$

Solution:

The parametric equation of a line through A is $\frac{x+5}{\cos\theta} = \frac{y+4}{\sin\theta} = r$

Let $AB = r_1, AC = r_2$ and $AD = r_3$

Then the coordinates of B, C, D are $(-5, +r_i\cos\theta - 4 + r_i\sin\theta), i = 1, 2, 3$

Now B lies on the line $x + 3y + 2 = 0 \therefore -5 + r_1\cos\theta + 3(-4 + r_1\sin\theta) + 2 = 0$

$\frac{15}{r_1} = \cos\theta + 3\sin\theta$

C lies on $2x + y + 4 = 0 \therefore 2(-5 + r_2\cos\theta) + (-4 + r_2\sin\theta) + 4 = 0$

$\Rightarrow \frac{10}{r_2} = 2\cos\theta + \sin\theta$

D lies on $x - y - 5 = 0 \therefore -5 + r_3\cos\theta + 4 - r_3\sin\theta - 5 = 0$

$\Rightarrow \frac{6}{r_3} = \cos\theta - \sin\theta$

From the given condition $\left(\frac{15}{r_1}\right)^2 + \left(\frac{10}{r_2}\right)^2 = \left(\frac{6}{r_3}\right)^2$

we get, $(\cos\theta + 3\sin\theta)^2 + (2\cos\theta + \sin\theta)^2 = (\cos\theta - \sin\theta)^2$

$\Rightarrow (2\cos\theta + 3\sin\theta)^2 = 0 \Rightarrow \tan\theta = -\frac{2}{3}$

$\therefore$ Equation of the line is

$y + 4 = -\frac{2}{3}(x + 5) \Rightarrow 2x + 3y + 22 = 0$

Q.3 Answer:

ay $-bx + 2b = 0$

Solution:

Slope of the line in the new position is $\frac{b}{a}$, since it is $\perp$ to the line $ax + by + c = 0$ and it cuts the x -axis at $(2, 0)$. Hence, the required line passes through $(2, 0)$ and its slope is $\frac{b}{a}$.

Required eq. is $y - 0 = \frac{b}{a}(x - 2) \Rightarrow ay = bx - 2b \Rightarrow ay - bx + 2b = 0$

Q.4 Answer:

$\frac{2\sqrt{2}}{3}$

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Solution:

The equation of the line through $(-1, 3)$ and having the slope 1 is $\frac{x+1}{\cos\theta} = \frac{y-3}{\sin\theta} = r$.

Any point on this line at a

distance r from $P(-1, 3)$ is

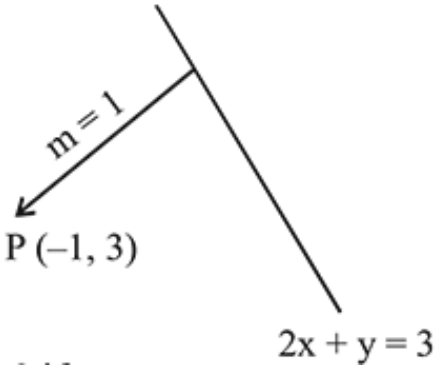
$(-1 + r\cos\theta, 3 + r\sin\theta)$ $\quad 2x + y = 3$

This point is on the line $2x + y = 3$ if $2(-1 + r\cos\theta) + 3 + r\sin\theta = 3$ ...(i)

But $\tan\theta = 1; \Rightarrow \theta = 45^\circ$

(i) becomes, $-2 + 2r \cdot \frac{1}{\sqrt{2}} + 3 + r \cdot \frac{1}{\sqrt{2}} = 3 \Rightarrow \frac{3r}{\sqrt{2}} = 2; \quad r = \frac{2\sqrt{2}}{3}$

Hence the required distance $= \frac{2\sqrt{2}}{3}$.



Q.5 Answer:

$-8x + 5y = 60$

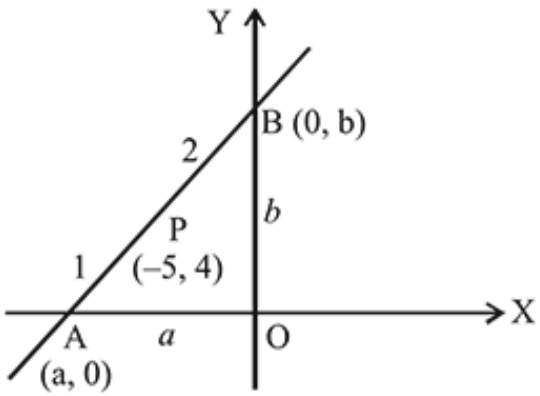
Solution:

(c) Let the equation of the line be $\frac{x}{a} + \frac{y}{b} = 1$.

So, the coordinates of A and B are $(a, 0)$ and $(0,\ b)$ respectively. Since the point $(-5, 4)$ divides AB in the ratio $1 : 2 \therefore \quad -5 = \frac{1.0+2\cdot a}{1+2}$ 13.

and $4 = \frac{1.b+2.0}{2+1} \Rightarrow a = -\frac{15}{2}$ and $b = 12$

So the line is $-\frac{2}{15}x + \frac{y}{12} = 1$, i.e. $-8x + 5y = 60$



Q.6 Answer:

$\frac{9}{(PR)^2} + \frac{16}{(PS)^2} = 1$

Solution:

The equation of the line in parametric form is

$\frac{x-3}{\cos\theta} = \frac{y-4}{\sin\theta} = r$

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Any point on this line is $(3 + r\cos\theta, 4 + r\sin\theta)$

It lies on $x = 6$ **if** $3 + r\cos\theta = 6 \Rightarrow r = 3\sec\theta$

$\therefore \mathrm{PR} = 3\sec\theta$

Again the point lies on $\mathrm{y} = 8$ **if** $4 + \mathrm{r}\sin\theta = 8$

$\therefore \quad \mathrm{r} = 4\,\mathrm{cosec}\,\theta$ **or** $\mathrm{PS} = 4\,\mathrm{cosec}\,\theta$

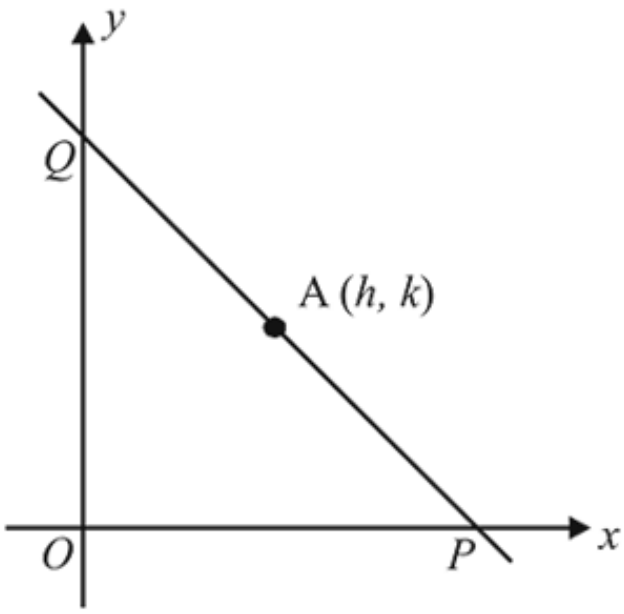
other options can be checked easily

Q.7 Answer:

$2hk$ sq. units

Solution:

(Let the equation of any line passing through $A(h, k)$ **be** $y - k = m(x - h)$**.**



Let this line cut the x **-axis and** y **-axis at** P **and** Q**. Then** $P \equiv \left(h - \frac{k}{m}, 0\right)$ **and** $Q \equiv (0, k - mh)$**.**

Let S **be the area of** ΔOPQ**, then** $S = \frac{1}{2}OP \times OQ = \frac{1}{2}\left(h - \frac{k}{m}\right)(k - mh)$

$= \frac{1}{2}\frac{(mh-k)(k-mh)}{m}$

$\Rightarrow 2mS = hkm - k^2 - h^2m^2 + khm$

$\Rightarrow h^2m^2 - 2(hk - S)m + k^2 = 0$

Since, m **is real,** $\therefore$ **its discriminant** $D \geq 0$

$\therefore 4(hk - S)^2 - 4h^2k^2 \geq 0$

$\Rightarrow S - 2hk \geq 0 \Rightarrow S \geq 2hk$

Hence, minimum value of S **is** $2hk$ **sq. units.**

Q.8 Answer:

8 points

Solution:

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$\left(x^2-a^2\right)^2\left(x^2-b^2\right)^2+c^4\left(y^2-a^2\right)^2=0$

This being the sum of two perfect squares, each term must be zero.

Hence, we get $\left(\mathrm{x}^2-\mathrm{a}^2\right)^2\left(\mathrm{x}^2-\mathrm{b}^2\right)^2=0$

or $\left(x^2-a^2\right)\left(x^2-b^2\right)=0$

or $(x-a)(x+a)(x-b)(x+b)=0$ **...(1)**

and $c^4\left(y^2-a^2\right)^2=0$

or $\mathrm{c}^2\left(\mathrm{y}^2-\mathrm{a}^2\right)=0$ **or** $\mathrm{c}^2(\mathrm{y}+\mathrm{a})(\mathrm{y}-\mathrm{a})=0 \ldots(2)$

Equation no. (1) holds good for $x=\pm a$ **or** $x=\pm b$ **Equation no.** (2) **is satisfied by** $y=\pm a$ **As both of these should be simultaneously satisfied, the given equation represents 8 points which we get as a result of different combinations of** (1) **and** (2)**, namely** $(\pm\mathrm{a}, \pm\mathrm{a}),(\pm\mathrm{b}, \pm\mathrm{a})$

Q.9 Answer:

$9x-20y+96=0$

Solution:

(a) Let the line cuts the axes at points $A(a, 0)$ **and** $B(0, b)$**. Now, given that** $(-4, 3)$ **divides** AB **in the ratio** $5:3$**. Then,** $-4=3a/8$ **and** $3=5b/8$**. Therefore,** $a=-32/3$ **and** $b=24/5$**. Then using the intercept form** $x/a+y/b=1$**, the equation of line is**

$-\frac{3x}{32}+\frac{5y}{24}=1$

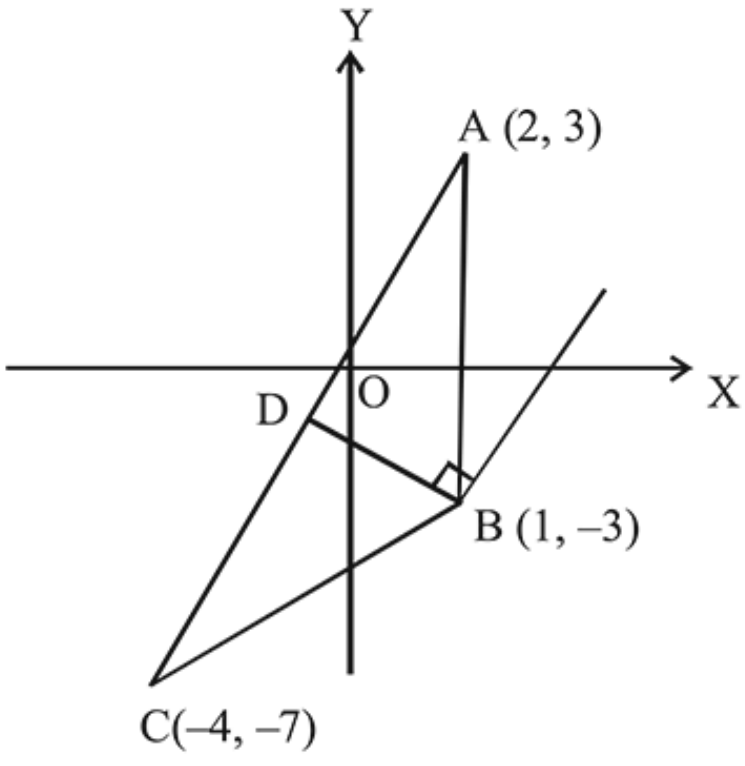
or $9x-20y+96=0$

Q.10 Answer:

$y-2x+5=0$

Solution:

Since, the line BD divides the triangle into two of equal area. BD is a median and Dis $(-1, -2)$**.**



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Slope of $\mathrm{BD} = -\frac{1}{2}$

So, the required line is $y + 3 = 2(x - 1) \Rightarrow \mathrm{y} - 2\mathrm{x} + 5 = 0$

Q.11 Answer:

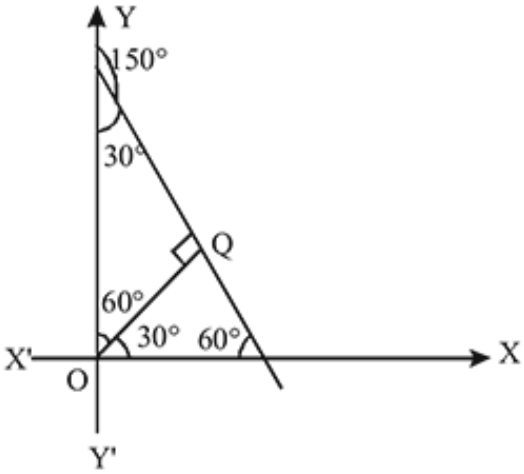
$\sqrt{3}x + y - 14 = 0$

Solution:

Here $p = 7$ **and** $\alpha = 30^\circ$

Equation of the

required line is



$x \cos 30^\circ + y \sin 30^\circ = 7$

or $\times \frac{\sqrt{3}}{2} + y \times \frac{1}{2} = 7$

or $\sqrt{3}x + y = 14$

Q.12 Answer:

$(0, 5)$

Solution:

The point of coincidence on the y **-axis is** $Q(0, \lambda)$

The image of $P(2, 3)$ **on the** y **-axis is** $P_1(-2, 3)$. P_1, Q **and** R **are collinear. Therefore, Slope of** $P_1Q =$ **Slope of** P_1R

or $\frac{\lambda - 3}{0 - (-2)} = \frac{10 - 3}{5 - (-2)}$

or $\lambda - 3 = 2$ **or** $\lambda = 5$.

Therefore, the point Q **is** $(0, 5)$.

Q.13 Answer:

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Solution:

Equation of line: $\frac{x}{a}+\frac{y}{b}=1$

Since line intersect coordinate axis at P **and** Q**, then**

$\frac{x}{OP}+\frac{y}{OQ}=1$

and line passes through $(8,2)$

So $\frac{8}{OP}+\frac{2}{OQ}=1$

$\therefore OP=\left(\frac{8OQ}{OQ-2}\right)$

So, $OP+OQ=\frac{8OQ}{OQ-2}+OQ$

To find its minimum value, we will differentiate the above expression

$\frac{d(OP+OQ)}{d(OQ)}$

$=\frac{(OQ-2)8-8OQ(1)+1}{(OQ-2)^2}$

$\Rightarrow 1=\frac{16}{(OQ-2)^2}$

$\Rightarrow (OQ-2)=\pm 4$

$\Rightarrow OQ=6$ **or** -2

So, $OQ=6$

$\therefore OP+OQ=18$

Q.14 Answer:

$(-24/5,17/5)$

Solution:

If three points $\mathrm{P}, \mathrm{A}, \mathrm{B}$ **form a triangle, then according to triangle inequalities,** $|\mathrm{PA}-\mathrm{PB}|<\mathrm{AB}$

If they do not form a triangle, i.e they are collinear, $|\mathrm{PA}-\mathrm{PB}|=\mathrm{AB}$

If they are collinear, then P **is the point of intersection of lines** $\mathrm{AB}, 4\mathrm{x}+3\mathrm{y}+9=0$ **(given** P **is on this line)**

equation of line $\mathrm{AB}=\mathrm{x}+2\mathrm{y}=2$

Solving the equations $4x+3y=-9$ **and** $x+2y=2$

$\Rightarrow 4\mathrm{x}+3\mathrm{y}-4(\mathrm{x}+2\mathrm{y})=-9-8$

$\Rightarrow 4\mathrm{x}+3\mathrm{y}-4\mathrm{x}-8\mathrm{y}=-17$

$\Rightarrow -5\mathrm{y}=-17$ or $\mathrm{y}=\frac{17}{5}$

Put $y=\frac{17}{5}$ **in** $x+2y=2$

$\Rightarrow \mathrm{x}=2-2\mathrm{y}=2-2\times\frac{17}{5}=2-\frac{34}{5}=\frac{10-34}{5}=\frac{-24}{5}$

point of intersection of the lines is $\left(\frac{-24}{5},\frac{17}{5}\right)$ **therefore** P **is** $\left(\frac{-24}{5},\frac{17}{5}\right)$.

Q.15 Answer:

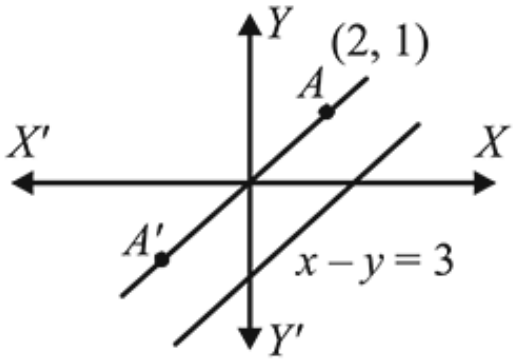
$(2-2\sqrt{2},1-2\sqrt{2})$

Solution:

Since the point $A(2,1)$ **is translated parallel to** $x-y=3$, AA' **has the same slope as that of** $x-y=3$**. Therefore,** AA' **passes through** $(2,1)$ **and has slope 1. Here,** $\tan\theta=1$ **or** $\cos\theta=$

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$1/\sqrt{2}, \sin\theta = 1/\sqrt{2}$.



Thus, the equation of AA' is

$\frac{x-2}{\cos(\pi/4)} = \frac{y-1}{\sin(\pi/4)}$

Since $AA' = 4$, the coordinates of A' are given by

$\frac{x-2}{\cos(\pi/4)} = \frac{y-1}{\sin(\pi/4)} = -4$

or $x = 2 - 4\cos\frac{\pi}{4}, y = 1 - 4\sin\frac{\pi}{4}$

or $x = 2 - 2\sqrt{2}, y = 1 - 2\sqrt{2}$

Hence, the coordinates of A' are $(2 - 2\sqrt{2}, 1 - 2\sqrt{2})$

Q.16 Answer:

$2Tx - a^2y + 2aT = 0$

Solution:

If the line cuts off the axes at A and B, then the area of triangle is

$\frac{1}{2} \times OA \times OB = T$

or $\frac{1}{2} \times a \times OB = T$ **or** $OB = \frac{2T}{a}$

Hence, the equation of line is

$\frac{x}{-a} + \frac{y}{2T/a} = 1$ **or** $2Tx - a^2y + 2aT = 0.$

Q.17 Answer:

$4x - 3y - 31 = 0$

Solution:

$3x + 4y + 5 = 0$ **or** $y = \frac{-3}{4}x + \frac{-5}{4}$

Slope $= \frac{-3}{4}$

Slope of required line, $\mathrm{m} = \frac{-1}{-\frac{3}{4}} = \frac{4}{3}$

Also line passes through $(4, -5)$

Equation of line, $y + 5 = \frac{4}{3}(x - 4) \Rightarrow 3y + 15 = 4x - 16$

$\Rightarrow 4x - 3y - 31 = 0$

Q.18 Answer:

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$(3, -1, 1)$

Solution:

Required plane is $A + \lambda B = 0$

Where $A = (x + y + z - 1)$ **and** $B = (2x + 3y - z - 4)$

That is,

$(x + y + z - 1) + \lambda(2x + 3y - z - 4) = 0$

$=> (1 + 2\lambda)x + (1 + 3\lambda)y + (1 - \lambda)z - (1 + 4\lambda) = 0$

As above equation is parallel to y**-axis, so** $\lambda = -1/3$

Therefore, required plane is $x + 4z + 1 = 0$

Substitute and check all options,

we get $(3, -1, 1)$ **is the point which lie in this plane.**

Q.19 Answer:

$2\alpha\beta$

Solution:

$\mathrm{P} \equiv (\alpha, \beta)$

Given line is $\frac{x}{a} + \frac{y}{b} = 1 \ldots . (1)$

If line (1) cuts x **and** y **axes at** A **and** B **respectively,**

then $\mathrm{A} \equiv (\mathrm{a}, 0)$ **and** $\mathrm{B} \equiv (0,\ \mathrm{b})$

Area of $\Delta \mathrm{OAB} = \mathbf{S}$

$\therefore \left|\frac{1}{2}ab\right| = \mathrm{S}$ **or** $\mathrm{ab} = 2\ \mathrm{S} \quad \ldots (2)$

Since line (1) passes through $\mathbf{P}(\alpha, \beta)$

$[\because ab > 0]$

$\frac{\alpha}{a} + \frac{\beta}{b} = 1$ **or,** $\frac{\alpha}{a} + \frac{a\beta}{2S} = 1$ **[from** $(2)]$

or, $a^2\beta - 2aS + 2\alpha S = 0$

Since a is real.

$\therefore 4\ \mathrm{S}^2 - 8\alpha\beta\mathrm{S} \geq 0$ **or** $4\ \mathrm{S}^2 \geq 8\alpha\beta\mathrm{S}$ **or** $\mathrm{S} \geq 2\alpha\beta$

Q.20 Answer:

$y = 2x$

Solution:

Obviously line through Q **is at greatest distance from point** P **when it is perpendicular to** PQ**. Now slope of line** PQ **is** $\mathrm{m_{PQ}} = -1/2$**. Then slope of**

or $2x - y = 0$

$y - 2 = 2(x - 1)$

perpendicular line passing through Q **is y=2x**

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